- What determines the value of an asset, whether it is real estate, an artwork, or a financial asset, such as shares in a firm?
- Markets for assets can work differently from markets for goods.
- Buyers have a secondary motive: not only to benefit in some way while owning the asset, but also to be able to sell it later.

- People buy assets for two reasons:
  1. To benefit from owning it
  2. To be able to sell it later
- Since the future is so uncertain, how should these assets be valued?
- The value of a financial asset (a **security**) depends on:
  1. the size of the cash flows that it is expected to generate.
  2. the uncertainty in one's forecasts of these cash flows.

Bond

- **Bond**: A security that promises to pay fixed amount of money at specific intervals.
- The risk of default on government bonds is usually negligible.
- When the stream of payments from an asset is fixed, the price of the asset will be inversely related to the interest rate it yields.
- Suppose the bond in consideration is a one-year bond and promises to pay 10,000 rupees a year from now.
- Let the price of the bond today by P rupees.
- If one were to buy the bond today, and hold it for a year then rate of return on holding the bond is $(10,000 - P)/P$.

- The interest rate on the bond is given by
$$i = \frac{10,000 - P}{P}.$$
- Example: If $P = 9000$, then the interest rate equals $1000/9000 = 11.1\%$ per year.
- Investors will be willing to buy and hold the asset only if its rate of return compares well with interest rates on similar assets elsewhere in the economy.
- If other interest rates in the economy rise, the interest rate on bonds will have to rise too — so the price of bonds will fall.

- Corporate bonds are not risk-free: high risk of default → high interest rate demanded by investors → lower bond price.
- The greater the risk of default, the higher will be the interest rate that investors demand in order to hold the asset.
- If two bonds promise exactly the same stream of payments, the riskier one will have a lower price.
- Investors will earn a higher interest rate if they buy the riskier bond and it happens not to default, but face a greater risk that the promised payments will not all materialize.

Stocks

- **Stocks** (shares) = A claim on a part of assets of a firm, and hence on its profits.
- Profits of a firm belong to its shareholders in direct proportion to the shares that each holds.
- These profits are typically divided between dividends paid directly to shareholders, and retained earnings used to maintain and expand the firm's ability to generate revenues.
- Stocks offer no specific promised stream of payments, and the time period over which payments will be made is not fixed.
- Firms expected to generate greater net earnings will have higher valuations — higher share price.

Risks

- Value of both bonds and stocks depends on uncertainty over its earnings.
- **Systematic risk**: Events that simultaneously affect broad classes of financial assets — Undiversifiable.
- Examples: changes in trade policy, interest rates, or economy-wide demand for goods and services.
- **Idiosyncratic risk**: Events that affect only a given firm/asset — Diversifiable and hence irrelevant for valuation.
- Example: successful development of a company product, or loss in lawsuit related to the company.

Risk (contd.)

- Investors will demand higher average returns from shares in companies with high levels of systematic risk, since their earnings will tend to be volatile in ways that cannot be easily diversified away.
- The rate of return that will induce investors to buy shares in a company is sometimes called the required rate of return, or the market capitalization rate.
- This rate will be higher for companies subject to greater systematic risk.
- So for any given beliefs about expected future earnings, share values will be higher for companies with lower market capitalization rates.

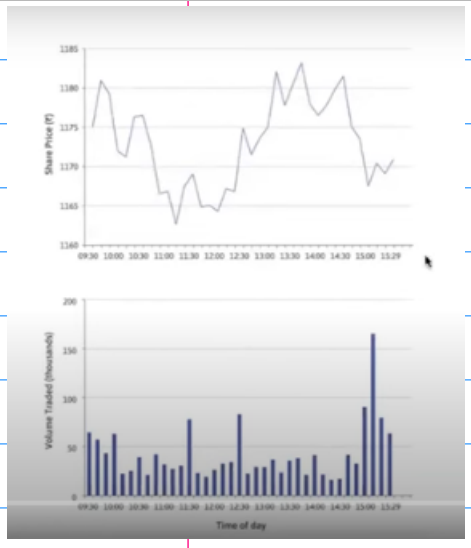
Trading Strategies.

- So far our analysis has focussed on determining the fundamental value of a share or security.
- Many institutional investors, including actively managed mutual funds and some hedge funds, adopt trading strategies based on buying assets that they perceive to be priced below their fundamental values, and selling those that are overpriced relative to fundamentals.

- There are other trading strategies that are not based on an assessment of fundamental value at all.
- For instance, some traders look for evidence of **momentum** in asset prices, buying because they expect prices to rise further, or selling because they expect prices to fall.
- An investor might be willing to pay more than the fundamental value of a security if she believes that the price will rise even further above this value.
- In this case, the buyer could make a gain by buying at a low price and selling at a higher price, even if the fundamental value of the security had not changed.

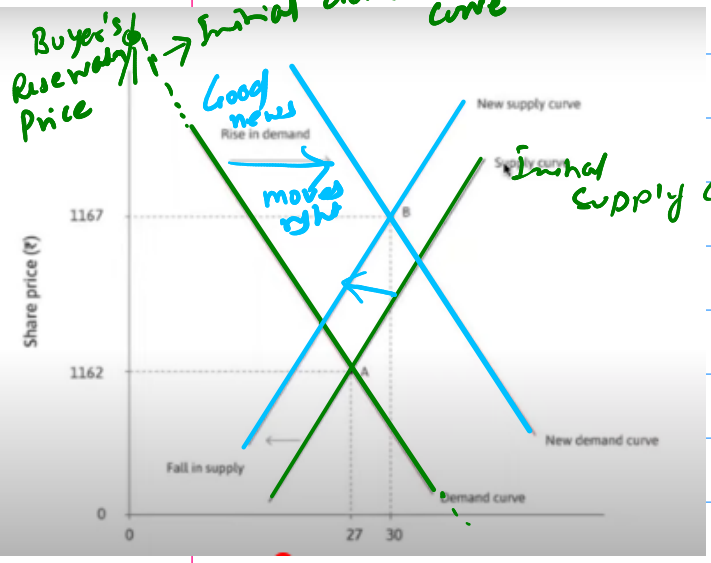
- Buying and selling assets based on an assessment of their fundamental values is a form of speculation, motivated by a belief that prices will soon return to their fundamental values.
- Buying and selling based on perceived momentum is also an example of speculation, motivated by a belief that short-run trends have a degree of persistence.
- These strategies, and many others, are present in modern financial markets.
- They determine the behaviour of prices, and the possibility of bubbles and crashes.

- When a company sells new shares (or stocks) to the public for the first time, this is called an initial public offering (IPO).
- After that, the shares are traded on the stock exchange.
- This is called trading on the secondary market.
- Stock exchange: A financial marketplace where shares (or stocks) and other financial assets are traded.



- Prices in trading in secondary markets are constantly changing.
- Previous graphs show how Reliance Industries Limited's (RIL) share price on the Sensex stock exchange fluctuated over one day in April 2020 and, in the lower panel, the number of shares traded at each point.

- At any time when the market for shares in RIL is open, each of the existing shareholders has a reservation price, namely the least price at which the shareholder would be willing to sell.
- Others are in the market to buy, as long as they can find an acceptable price.
- As traders' beliefs about the profitability of RIL change, their willingness to buy and sell changes.



Imp. When share price ↑ demand curve moves to right & supply curve moves to left & Vice - Versa.

- Anyone wishing to buy/sell can submit a price and quantity combination known as a **limit order**.
- For instance, a limit order to buy 1,000 shares in RIL at a price of Rs. 1,162 per share indicates that the buyer commits to buying 1,000 shares, as long as they can be obtained at a price no greater than Rs. 1,162 per share.
- This is the buyer's reservation price.
- A limit sell order indicates a commitment to sell a given quantity of shares, as long as the price is no less than the amount specified (the seller's reservation price).

- When a limit buy order is placed, one of two things can happen.
- If a previously placed limit sell order exists that has not yet been filled, and it offers the required number of shares at a price that is at or below the amount indicated by the buyer, a trade occurs.
- If there is no such order available, then the limit order is placed in what is called an **order book** (which is really just an electronic record), and becomes available to trade against new sell orders that arrive.

- Orders to buy are referred to as **bids**, and orders to sell as **asks**.
- The order book lists bids in decreasing order of price, and asks in increasing order.

Bid (Buyer) Ask (seller)

Bid		Ask	
Price (Rs.)	Quantity	Price (Rs.)	Quantity
1,418.05	15	1,418.20	50
1,417.90	240	1,418.25	1
1,417.85	500	1,418.30	1,686
1,417.80	433	1,418.35	1
1,417.60	459	1,418.40	2

- Given this situation, a buy order for 100 shares at Rs. 1,418.10 would remain unfilled and would enter the book at the top of the bid column.
- However, a bid for 1,500 shares at Rs. 1,418.30 would be filled immediately, since it can be matched against existing limit sell orders.
- 50 shares would trade at Rs. 1,418.20 apiece, 1 share would trade at Rs. 1,418.25, and the rest would trade at Rs. 1,418.30.

1418.20 → 50.
1418.25 → 1.
1418.30 → 1449

- Whenever a buy order is immediately filled, trade occurs at the best possible price for the buyer (the ask price).
- Similarly, if a sell order is placed and immediately filled from existing orders, trade occurs at the best possible price for the seller (the bid price).
- How do prices in such a market change over time?

due to good/bad news or his buyer/seller.

- Someone who receives negative news about RIL, such as a rumour that a board member is about to resign, and believes this information has not yet been incorporated into the price, may place a large sell order at a price below Rs. 1,418.05, which will immediately trade against existing bids.
- As these trades occur, bids are removed from the order book and the price of the stock declines.
- Similarly, in response to good news, orders to buy at prices above the lowest ask will trade against existing sell orders, and transactions will occur at successively increasing prices.

- Since the price fluctuates, it is not easy to think of this market as being in equilibrium.
- But it is nevertheless the case that the price is always adjusting to reconcile supply and demand and hence clear the market.
- Those who believe they will benefit from buying RIL shares at a particular price lodge a bid at that price.
- Those who believe they will benefit by selling lodge an ask at a particular price.
- The price at any moment reflects the aggregate outcome of the behaviour of all the actors in the market—including those who are simply holding on to their shares → Imp. to remember.

- Share prices are not only volatile hour-by-hour and day-by-day.
- They can also display large swings.
- Example: the value of the Nasdaq Composite Index between 1995 and 2004.
- This index is an weighted average of prices for a set of stocks.
- The Nasdaq Composite Index at this time included many fast-growing and hard-to-value companies in technology sectors.

NASDAQ (Jan 95 - Jan 04)



- The index began the period at less than 750, and rose in five years to more than 5,000 with a remarkable annualized rate of return of around 45%.
- It then lost two-thirds of its value in less than a year, and eventually bottomed out at around 1,100, almost 80% below its peak.
- The episode has come to be called the **tech bubble**.

- The term **bubble** refers to a sustained and significant departure of the price of any asset (financial or otherwise) from its fundamental value.
- Sometimes, new information about the fundamental value of an asset is quickly and reliably expressed in markets.
- Changes in beliefs about a firm's future earnings growth result in virtually instantaneous adjustments in its share price.
- Both good and bad news about patents or lawsuits, the illness or departure of important personnel, earnings surprises, or mergers and acquisitions can all result in active trading—and swift price movements.

- Because stock price movements often reflect important information about the financial health of a firm, traders who lack this information can try to deduce it from price movements.
- Changes in prices are messages containing information.
- If markets are to work well, traders must respond to these messages.
- But when they interpret a price increase as a sign of further price increases (momentum trading strategies) the result can be self-reinforcing cycles of price increases, resulting in asset price bubbles followed by sudden price declines, called crashes.

Imp. Correlate with present Indian conditions.

- Three distinctive and related features of markets may give rise to bubbles:
  1. Resale value: Assets offer the opportunity for speculation on a change in its price. Eg: houses, paintings.
  2. Ease of trading: In financial markets, the ease of trading means that it is easier to switch between being a buyer and being a seller.
  3. Ease of borrowing to finance purchases: If market participants can borrow to increase their demand for an asset that they believe will increase in price, this allows an upward movement of prices to continue.

- Asset prices can swing wildly, giving rise to the impression that they bear little relation to the stream of income that might reasonably be expected from holding them.
- But do bubbles really exist, or are they an illusion based only on hindsight?
- In other words, is it possible to know that a market is experiencing a bubble before it crashes?

→ Example. huge increase in house's prices (not correlating with the demand)

→ 2 Thoughts. Yes / No.

Nobel prize winner. No

- Eugene Fama: "These words have become popular. I don't think they have any meaning ... It's easy to say prices went down, it must have been a bubble, after the fact. I think most bubbles are twenty-twenty hindsight. Now after the fact you always find people who said before the fact that prices are too high. People are always saying that prices are too high. When they turn out to be right, we anoint them. When they turn out to be wrong, we ignore them. They are typically right and wrong about half the time."

Argument.

- Efficient market hypothesis: all generally available information about fundamental values is incorporated into prices virtually instantaneously.
- "If an economist had a formula that could reliably forecast crises a week in advance, say, then that formula would become part of generally available information and prices would fall a week earlier."

- Counterargument: "In practice each individual investor does not know when other investors will start trading against the bubble. This uncertainty makes each individual investor nervous about whether he can be out of (or short) the market sufficiently long until the bubble finally bursts. Consequently, each investor is reluctant to lean against the wind. Indeed, investors may in fact prefer to ride a bubble for a long time such that price corrections only occur after a long delay, and often abruptly."
- Citibank CEO (2007): "When the music stops, ..., things will be complicated. But as long as the music is playing, you've got to get up and dance. We're still dancing."

Yes.

- Robert Shiller has argued that relatively simple and publicly observable statistics, such as the ratio of stock prices to earnings per share, can be used to identify bubbles as they form.
- Leaning against the wind by buying assets that are cheap based on this criterion, and selling those that are dear, can result in losses in the short run, but long-term gains that, in Shiller's view, exceed the returns to be made by simply investing in a diversified basket of securities with similar risk attributes.

Take-away.

- Fama's view: Asset prices throughout the (tech bubble) episode were based on the best information available at the time and fluctuated because information about the prospects of the companies was changing sharply.
- Shiller's view: Prices in the late 1990s had been driven up simply by expectations that the price would still rise further. He called this '**irrational exuberance**' among investors.

Financial Crisis - 2008.

Housing Prices



- Analysts anticipated that the decline in housing prices starting in 2006, would lead to a decrease in demand and a slowdown in growth
- Few anticipated that it would lead to a major macroeconomic crisis.
- Key missing analysis was in terms of thinking about the effect on the financial system and the economy.

- Price increase during the early 2000s was seen emerging as a result of interest rate environment and innovations in financial products.
- Interest rates in early 2000s had come down a lot - these translated in a low mortgage rate which increased the demand for housing and th the housing price.
- The development of products such as **subprime mortgages** allowed expansion of loans to more risky borrowers.

- As long as housing prices kept increasing, such developments were not considered problematic.
- However prices did start going down from 2006 - which resulted in the value of mortagages exceeding the value of their house.
- The mortagages also turned out to be riskier that what borrowers ha of understood them to be.
- Many borrowers ended up defaulting.

(1) (2) (3).

- High leverage, illiquidity of assets and liquidity of liabilities all increased risk in the financial system.
- In 2008, all three elements were present in the financial system.
- Banks were highly leveraged: ✓
- They underestimated the risk. ✓
- The mangers were incentivized to take up high expected returns projects.
- Banks were able to sidestep regulation through **structured investment vehicles** (SIVs) (SIVs)

- On the liability side: SIVs borrowed from investors, in the form of short term debt.
- On the asset side: SIVs held various type of securities.
- SIVs had a guarantee from the bank that created them that if neede the bank would provide funds to SIV.
- SIVs allowed banks to get around regulatory requirements regarding leverage.
- Fall in price of houses and defaults on mortgage loans lead to questions about solvency of SIVs and banks backing them. ✓

Imp. to understand.

- Another development was the growth of securitization.
- If banks only made loans locally, a fall in house prices locally would lead to banks getting in trouble.
- If banks had portfolios consisting of mortgages from different parts of the country, these banks might avoid getting into trouble.
- Example of such a product was **mortgage-backed security**.
- Such securities would lead to banks paying less attention to whether borrower could actual repay, and also lead to rating agencies incorrectly assessing the value of such securities.

- When house prices started declining, investors became reluctant to hold them or to continue lending to SIVs which held them.
- Many of the assets held by banks and SIVs were thus illiquid.
- They were extremely hard to assess and to sell except at fire sale prices.

- Another development was that of **wholesale funding**.
- Banks relied on funding from other banks or other investors in the form of short term debt.
- SIVs were entirely funded through such wholesale funding.
- Holders of checkable deposits were protected by deposit insurance, but not other investors.
- When investors became worried about the value of assets held by SIVs, they started asking for their funds back.

Remember This.

1. 2. 3.

- Combination of high leverage, liquid liabilities and illiquid assets led to the financial crisis.
- High leverage led to sharp decline in capital of banks and SIVs.
- This led them to sell some of their assets.
- Since assets were complex, firms had to sell them at firesale prices.
- This lead to further decrease in their assets.
- Investors also became reluctant to lend money to such SIVs and banks.

- Large increase in interest rate at which people and firms could borro
- Large reduction in confidence.
- Result of lower confidence, lower stock prices and lower housing pric led to a reduction in consumption.
- The fall in consumption and investment resulted in fall in aggregate demand.

- Federal deposit insurance was expanded.
- Federal Reserve provided liquidity to the financial system which mad it easier for banks and other financial intermediaries to borrow from the Fed.
- Government introduced the **Troubled Asset Relief Program (TARP)** with an aim to remove the complex assets from the bank balance sheet.
- This fund was later on used to increase the capital of banks.

- Fed also started to reduce the policy rate.
- By December 2008, the rate was down to zero.
- Thus monetary policy was constrained by the zero lower bound.
- The Fed thus had to resort to unconventional monetary policy.
- The government also resorted to **fiscal policy** - reduction in taxes ar increases in government spending - with an aim to increase demand.

- One of the remarkable stories in the lead up to the Global Financial Crisis was the dramatic rise and fall of structured finance products.
- In under a decade, issuance of these products within the U.S. economy grew more than ten-fold.
- In the first three quarters of 2005, $25–$40 billion of structured finance products were issued in each quarter.
- In the last quarter of 2006 and the first two quarters of 2007, issuance of structured finance products peaked at about $100 billion in each quarter.
- But by the first two quarters of 2008, these quantities had dropped to less than $5 billion per quarter.

- Structured finance activities is the pooling of economic assets like loans, bonds, and mortgages, and the subsequent issuance of a prioritized capital structure of claims, known as tranches, against these collateral pools.
- Because of the prioritization scheme used in structuring claims, many of the manufactured tranches are far safer than the average asset in the underlying pool.

- This ability of structured finance to repackage risks and to create "safe" assets from otherwise risky collateral led to a dramatic expansion in the issuance of structured securities, most of which were viewed by investors to be virtually risk-free and certified as such by the rating agencies.
- At the core of the financial market crisis of 2008 was the discovery that these securities were actually far riskier than originally advertised.

- How did the process of securitization allow trillions of dollars of risky assets to be transformed into securities that were widely considered to be safe?
- Most securities could only have received high credit ratings if the rating agencies were extraordinarily confident about their ability to estimate the underlying securities' default risks, and how likely defaults were to be correlated.
- Modest imprecision in the parameter estimates can lead to variation in the default risk of the structured finance securities that is sufficient, for example, to cause a security rated AAA to default with reasonable likelihood.

- A second, equally neglected feature of the securitization process is that it substitutes risks that are largely diversifiable for risks that are highly systematic.
- As a result, securities produced by structured finance activities have far less chance of surviving a severe economic downturn than traditional corporate securities of equal rating.
- Moreover, because the default risk of senior tranches is concentrated in systematically adverse economic states, investors should demand far larger risk premia for holding structured claims than for holding comparably rated corporate bonds.

- Both of these features of structured finance products - the extreme fragility of their ratings to modest imprecision in evaluating underlying risks and their exposure to systematic risks - go a long way in explaining the spectacular rise and fall of structured finance.

- For over a century, agencies such as Moody's, Standard and Poor's, and Fitch have gathered and analyzed a wide range of financial, industry, and economic information to arrive at independent assessments on the creditworthiness of various entities, giving rise to the widely popular rating scales (AAA, AA, A, BBB, and so on).
- The agencies historically focused the majority of their business on single-name corporate finance — that is, issues of creditworthiness of financial instruments that can be clearly ascribed to a single company.
- In the years leading up to the crisis, the business model of credit rating agencies had expanded beyond their historical role to include the nascent field of structured finance.

- From its beginnings, the market for structured securities evolved as a "rated" market, in which the risk of tranches was assessed by credit rating agencies.
- Issuers of structured finance products were eager to have their new products rated on the same scale as bonds so that investors subject to ratings-based constraints would be able to purchase the securities.
- By having these new securities rated, the issuers created an illusion of comparability with existing "single-name" securities.
- This provided access to a large pool of potential buyers for what otherwise would have been perceived as very complex derivative securities.

- Risks of all kinds have been repackaged to create vast quantities of triple-A–rated securities with competitive yields.
- By mid-2007, there were 37,000 structured finance issues in the U.S. alone with the top rating (Scholtes and Beales, 2007).
- According to Fitch Ratings (2007), roughly 60 percent of all global structured products were AAA-rated, in contrast to less than 1 percent of the corporate issues.
- By offering AAA-ratings along with attractive yields during a period of relatively low interest rates, these products were eagerly bought up by investors around the world.

- Structured finance activities grew to represent a large fraction of Wall Street and rating agency revenues in a relatively short period of time.
- By 2006, structured finance issuance led Wall Street to record revenue and compensation levels.
- The same year, Moody's Corporation reported that 44 percent of its revenues came from rating structured finance products.

- By 2008, everything had changed.
- Global issuance of collateralized debt obligations slowed to a crawl and Wall Street banks were forced to incur massive writedowns.
- Rating agency revenues from rating structured finance products disappeared virtually overnight and the stock prices of these companies fell by 50 percent, suggesting the market viewed the revenue declines as permanent.

- A huge fraction of existing products saw their ratings downgraded.
- Downgrades were particularly widespread among what are called "asset-backed security" collateralized debt obligations—which are comprised of pools of mortgage, credit card, and auto loan securities.
- For example, 27 of the 30 tranches of asset-backed collateralized debt obligations underwritten by Merrill Lynch in 2007 saw their triple-A ratings downgraded to "junk".
- Overall, in 2007, Moody's downgraded 31 percent of all tranches for asset-backed collateralized debt obligations it had rated and 14 percent of those initially rated AAA.
- By mid-2008, structured finance activity was effectively shut down.

Manufacturing AAA Rated Securities

- Manufacturing securities of a given credit rating requires tailoring the cashflow risk of these securities to satisfy the guidelines set forth by the credit rating agencies.
- Cashflow risk is measured by the likelihood of default and the magnitude of loss incurred in the event of a default.
- Structured finance allows originators to accomplish this goal by means of a two-step procedure involving **pooling** and **tranching**.

- In the first step, a large collection of credit-sensitive assets is assembled in a portfolio, which is typically referred to as a "special purpose vehicle".
- The special purpose vehicle is separate from the originator's balance sheet to isolate the credit risk of its liabilities — the tranches — from the balance sheet of the originator.

- If the special purpose vehicle issued claims that were not prioritized and were simply fractional claims to the payoff on the underlying portfolio, the structure would be known as a pass-through securitization.
- At this stage, since the expected portfolio loss is equal to the mean expected loss on the underlying securities, the portfolio's credit rating would be given by the average rating of the securities in the underlying pool.
- The pass-through securitization claims would inherit this rating, thus achieving no credit enhancement.

- To manufacture a range of securities with different cash flow risks, structured finance issues a capital structure of prioritized claims, known as tranches, against the underlying collateral pool.
- The tranches are prioritized in how they absorb losses from the underlying portfolio.

- Example: senior tranches only absorb losses after the junior claims have been exhausted, which allows senior tranches to obtain credit ratings in excess of the average rating on the average for the collateral pool as a whole.
- The degree of protection offered by the junior claims, or overcollateralization, plays a crucial role in determining the credit rating for a more senior tranche, because it determines the largest portfolio loss that can be sustained before the senior claim is impaired.

- This process of pooling and tranching can be illustrated with a two-asset example.
- Consider two identical securities ("bonds") both of which have a probability of default p_D, and pay 0 conditional on default and 1 otherwise.
- Suppose we pool these securities in a portfolio, such that the total notional value of the underlying fund is 2, and then issue two tranches against this fund, each of which pay 1.

- A "junior" tranche can be written such that it bears the first 1 of losses to the portfolio; thus, the junior tranche pays 1 if both bonds avoid default and zero if either bond defaults.
- The second, "senior" claim, which bears losses if the capital of the junior tranche is exhausted, pays 1 if neither bond defaults or if only one out of two bonds defaults; it only defaults if both bonds default.

- To compute the expected cash flows (or default probabilities) for the tranches, we will need to know the likelihood of observing both bonds defaulting simultaneously.
- In this example, the default dependence structure can be succinctly described by means of a single parameter — either the joint probability of default, or the default correlation.

- If the defaults of the two bonds are imperfectly correlated, the senior tranche will pay either 1 or 0 — just like the individual bonds — except that it will be less likely to default than either of the underlying bonds.
- Example: If the two bonds have a 10 percent default probability and defaults are uncorrelated, the senior tranche will only have a 1 percent chance of default.

- This basic procedure allows highly risky securities to be repackaged, with some of the resulting tranches sold to investors seeking only safe investments.
- Junior tranches, being risky, will have low prices and high promised returns.
- Senior tranches, being relatively safe, will have relatively higher prices and lower promised returns.

$P(\text{default both}) = 0.1 \times 0.1 = 0.01$

$P(\text{No default}) = 0.9 \times 0.9 = 0.81$

$P(\text{one default}) = 1 - 0.81 - 0.01 = 0.18.$

$\therefore P(\text{Senior Tranch Safe}) = 0.01$ i.e. $\therefore$ higher ratings (else its risk was 0.1)

- Central insight of structured finance: by using a larger number of securities in the underlying pool, a progressively larger fraction of the issued tranches can end up with higher credit ratings than the average rating of the underlying pool of assets.
- Example: Extend the two-bond example by adding a third bond that pays 0 or 1.
- Three claims can be issued against this underlying capital structure.

- First tranche defaults if any of the three bonds default, the second tranche defaults if two or more of the bonds default, and the final, senior-most tranche only defaults when all three bonds default.
- If bonds default 10 percent of the time and defaults are uncorrelated, the senior tranche will now default only 0.1 percent of the time, the middle tranche defaults 2.8 percent of the time, and the junior tranche defaults 27.1 percent of the time.
- Thus, by including a third bond in the pool, two-thirds of the capital — as measured by the tranche notional values — can be repackaged into claims that are less risky than the underlying bonds.

3 tranches (senior, middle, junior)

senior defaults if all 3 default.

$P(S) = 0.1 \times 0.1 \times 0.1 = 0.001$

$= 0.1\%$

middle defaults (if 2 tranches default + when 3 tranches default) : $0.001 + 3 \times 0.1 \times 0.1^2 \times 0.9$
$= 0.001 + 0.027 = 0.028$
$= 2.8\%$

Junior defaults when either 1 defaults or 2 defaults or 3 defaults.
$= 3 \times 0.1 \times 0.9 \times 0.9 + 0.027 + 0.001 = 0.243 + 0.028 = 0.271 = 27.1\%$.

- Another way to increase the total notional value of highly-rated securities produced is to reapply the securitization machinery to the junior tranches created in the first round.
- For example, in the two-bond case in which defaults were uncorrelated, the 1 junior tranche defaults with 19 percent probability.
- However, if we combine this 1 junior tranche with an identical 1 junior tranche created from another two-bond pool, we can again tranche the resulting 2 of capital into two prioritized 1 claims.

- If there continues to be no correlation among underlying assets, the resulting senior tranche from this second round of securitization - a tranche that defaults if at least one bond defaults in each of the two underlying pools - has a default probability of 3.6 percent.
- The collateralized debt obligations created from the tranches of other collateralized debt obligations are typically called CDO-squared — that is, CDO^2.

- A key factor determining the ability to create tranches that are safer than the underlying collateral is the extent to which defaults are correlated across the underlying assets.
- The lower the default correlation, the more improbable it is that all assets default simultaneously and therefore the safer the senior-most claim can be made.
- Conversely, as bond defaults become more correlated, the senior-most claims become less safe.

- For example, consider the two-bond case in which defaults are perfectly correlated.
- Since now both bonds either survive or default simultaneously, the structure achieves no credit enhancement for the senior tranche.
- Thus, in the two-bond example, while uncorrelated risks of default allow the senior claim to have a 1 percent default probability, perfectly correlated risks of default would mean that the senior claim inherits the risk of the underlying assets, at 10 percent.
- Finally, intermediate levels of correlation allow the structure to produce a senior claim with default risk between 1 and 10 percent.

Main takeaways : In securitization, by coupling (pooling & tranching) Senior gets higher ratings.

Your portfolio consists of two green bonds: each bond has a face value 100 and a probability of default $p = 10\%$. However, both green bonds have a correlation $\gamma \in [-1, 1]$. Through securitization, you are able to create two new bonds with face values: a senior bond (the "yellow bond"), and a junior bond (the "red bond"). We are interested in calculating the fair value of the yellow and red bonds.

Lets solve step by step

Problem 1A

First, we assume that the green bonds are independent: in this case, $\gamma = 0$.

1. What is the probability that both green bonds default?
2. What is the probability that none of the green bonds default?
3. What is the probability that exactly one of the green bonds defaults?
4. Your portfolio of green bonds has the following values: [1] 200 if none of the green bonds default, [2] 100 if exactly one green bond defaults, [3] 0 if both green bonds default. What is the expected value of your portfolio?
5. What is the standard deviation of your portfolio?
6. Focus on the yellow bond's payoff in each possible state of the world. What will be the payoff of the yellow bond if none of the green bonds default? What will be the payoff of the yellow bond if exactly one of the green bonds default? What will be the payoff of the yellow bond if both the green bonds default?
7. Focus on the red bond's payoff in each possible state of the world. What will be the payoff of the red bond if none of the green bonds default? What will be the payoff of the red bond if exactly one of the green bonds default? What will be the payoff of the red bond if both the green bonds default?
8. What is the expected payoff of a yellow bond across all states?
9. What is the expected payoff of a red bond across all states?

Your portfolio consists of two green bonds: each bond has a face value 100 and a probability of default $p = 10\%$. However, both green bonds have a correlation $\gamma \in [-1, 1]$. Through securitization, you are able to create two new bonds with face values: a senior bond (the "yellow bond"), and a junior bond (the "red bond"). We are interested in calculating the fair value of the yellow and red bonds.

Problem 1A

(uncorrelated)

First, we assume that the green bonds are independent: in this case, $\gamma = 0$.

1. What is the probability that both green bonds default? $= 0.1 \times 0.1 = 0.01 = 1\%$
2. What is the probability that none of the green bonds default? $= 0.9 \times 0.9 = 0.81 = 81\%$
3. What is the probability that exactly one of the green bonds defaults? $= 2 \times 0.1 \times 0.9 = 0.18$
4. Your portfolio of green bonds has the following values: [1] 200 if none of the green bonds default, [2] 100 if exactly one green bond defaults, [3] 0 if both green bonds default. What is the expected value of your portfolio? $E(V) = 0.01 \times 0 + 0.81 \times 200 + 0.18 \times 100 = 162 + 18 = 180$
5. What is the standard deviation of your portfolio?

$\therefore E(X^2) - [E(X)]^2 = 34200 - (180)^2 = 34200 - 32400$

$[E(X^2) = E(200^2) = 0.01 \times 0 + 0.81 \times 40000 + 0.18 \times 10000$
$= 0 + 32400 + 1800 = 34200]$

$\therefore$ Variance : $34200 - 32400 = 1800$
$\therefore$ Standard deviation : $\sqrt{1800}$

6. Focus on the yellow bond's payoff in each possible state of the world. What will be the payoff of the yellow bond if none of the green bonds default? What will be the payoff of the yellow bond if exactly one of the green bonds default? What will be the payoff of the yellow bond if both the green bonds default?

Yellow bond is Senior bond.
P(default of senior bond) $= 0.1 \times 0.1 = 0.01$
∴ Payoff may be $100 - 0.01 \times 100 = 100 - 1 = 99$

payoff if none defaults = 100
payoff if only one defaults = 100
payoff if both default = 0.

7. Focus on the red bond's payoff in each possible state of the world. What will be the payoff of the red bond if none of the green bonds default? What will be the payoff of the red bond if exactly one of the green bonds default? What will be the payoff of the red bond if both the green bonds default?

payoff (red bond) if none defaults $= 0.81 \times 100 = 81$
payoff if one defaults = 0
payoff if both default = 0.

8. What is the expected payoff of a yellow bond across all states? $= 0.01 \times 0 + 0.81 \times 100 + 0.18 \times 100$
$= 81 + 18 = 99$

9. What is the expected payoff of a red bond across all states?

= red bond gets only when no one defaults.
$= 0.81 \times 100 = 81$

Now, we assume that the green bonds are perfectly correlated: in this case, $\gamma = 1$.

(Now we have to consider both as single entity) with Payoff = 200 & Pd = 0.1

1. What is the probability that both green bonds default? = 0.1
2. What is the probability that none of the green bonds default? 0.9
3. What is the probability that exactly one of the green bonds defaults? = 0
4. Your portfolio of green bonds has the following values: [1] 200 if none of the green bonds default, [2] 100 if exactly one green bond defaults, [3] 0 if both green bonds default. What is the expected value of your portfolio? $= 0.9 \times 200 + 0.1 \times 0$
$= 180$
5. What is the standard deviation of your portfolio? $V = E(X^2) - E(X)^2 = 0.9 \times 40000 - 180^2 = 36000 - 32400 = 3600$
$\therefore SD = \sqrt{3600}$
6. Focus on the yellow bond's payoff in each possible state of the world. What will be the payoff of the yellow bond if none of the green bonds default? What will be the payoff of the yellow bond if exactly one of the green bonds default? What will be the payoff of the yellow bond if both the green bonds default?
= 100, if anyone defaults
= 0 if both default.
7. Focus on the red bond's payoff in each possible state of the world. What will be the payoff of the red bond if none of the green bonds default? What will be the payoff of the red bond if exactly one of the green bonds default? What will be the payoff of the red bond if both the green bonds default?
if both default = 0
if one defaults = 0
if no one defaults = 100
8. What is the expected payoff of a yellow bond across all states? → $0.9 \times 100 = 90$
9. What is the expected payoff of a red bond across all states? $= 0.9 \times 100 = 90$
10. Compare the expected payoffs you found in parts 8 and 9 with the corresponding ones you found for $\gamma = 0$.

Challenges of Rating Structured Products

- Credit ratings are designed to measure the ability of issuers or entities to meet their future financial commitments, such as principal or interest payments.
- Rating is either based on the anticipated likelihood of observing a default, or it is based on the expected economic loss: the product of the likelihood of observing a default and the severity of the loss conditional on default.
- A credit rating can intuitively be thought of as a measure of a security's expected cash flow.

- In the context of corporate bonds, securities rated BBB or higher have come to be known as investment grade and are thought to represent low to moderate levels of default risk.
- Those rated BB+ and below are referred to as speculative grade and are already in default or closer to it.

Table 1

Historical Default Experience of Bonds Rated by Fitch

	Investment-grade Bonds									
Rating at issuance	*AAA*	*AA+*	*AA*	*AA−*	*A+*	*A*	*A−*	*BBB+*	*BBB*	*BBB−*
10-year default probability	0.19%	0.57%	0.89%	1.15%	1.65%	1.85%	2.44%	3.13%	3.74%	7.26%
Default rate (annualized)	0.02%	0.06%	0.09%	0.12%	0.17%	0.19%	0.25%	0.32%	0.38%	0.75%

	Speculative-grade Bonds									
Rating at issuance	*BB+*	*BB*	*BB−*	*B+*	*B*	*B−*	*CCC+*	*CCC*	*CC*	*C*
10-year default probability	10.18%	13.53%	18.46%	22.84%	27.67%	34.98%	43.36%	48.52%	77.00%	95.00%
Default rate (annualized)	1.07%	1.45%	2.04%	2.59%	3.24%	4.30%	5.68%	6.64%	14.70%	29.96%

Imp. – In Investment grade bond, max default rate – 7.26% or 0.75%.

whereas in Speculative best is 10.18% or 1.07%

& Worst is 95.% or 29.96%

- In the single-name rating business, where the credit rating agencies had developed their expertise, securities were assessed independently of each other, allowing rating agencies to remain agnostic about the extent to which defaults might be correlated.
- To assign ratings to structured finance securities, the rating agencies were forced to address the bigger challenge of characterizing the entire joint distribution of payoffs for the underlying collateral pool.
- Riskiness of collateralized debt obligation tranches is sensitive to the extent of commonality in default among the underlying assets

- The structure of collateralized debt obligations magnifies the effect of imprecise estimates of default likelihoods, default correlation, as well as model errors due to the potential misspecification of default dependencies.
- These problems are further increased when different financial securities are combined together, as in CDO-squared.
- With multiple rounds of structuring, even minute errors at the level of the underlying securities that would be insufficient to alter the security's rating can dramatically alter the ratings of the structured finance securities.

Challenge of Rating Structured Finance Assets

- To illustrate the sensitivity of the collateralized debt obligations and associated securities such as CDO-squared to errors in parameter estimates, a simulation exercise can be conducted.
- Payoffs to 40 collateralized debt obligation pools are simulated, each comprised of 100 bonds with a five-year default probability of 5 percent and a recovery rate of 50 percent of face value conditional on default.
- Using the historical default experience data, each bond in the collateral pool would garner a just-below investment grade rating of BB+.
- Pairwise bond default correlation is set at 0.20 within each collateral pool and it is assumed the defaults of bonds belonging to different collateral pools are uncorrelated.

Challenge of Rating Structured Finance Assets

- Within each collateral pool, a capital structure is constructed comprised of three tranches prioritized in order of their seniority.
- The "junior tranche" is the first to absorb losses from the underlying collateral pool and does so until the portfolio loss exceeds 6 percent, at which point the junior tranche becomes worthless.
- The "mezzanine tranche" begins to absorb losses once the portfolio loss exceeds 6 percent and continues to do so until the portfolio loss reaches 12 percent.
- Finally, the senior tranche absorbs portfolio losses in excess of 12 percent.
- A CDO-squared is also constructed by issuing a second capital structure of claims against a pool that combines the mezzanine tranches from the 40 original collateralized debt obligations.

Table 2

Summary Statistics for CDO and CDO^2 Tranches in our Simulation under Baseline Parameters

	Attachment points	*Default probability*	*Expected payoff*	*Rating*
CDO				
Junior	0%–6%	97.52%	0.59	NR
Mezzanine	6%–12%	2.07%	> 0.99	BBB−
Senior	12%–100%	< 0.00%	> 0.99	AAA
CDO^2 ([6, 12])				
Junior	0%–6%	56.94%	0.93	C
Mezzanine	6%–12%	< 0.00%	> 0.99	AAA
Senior	12%–100%	< 0.00%	> 0.99	AAA

Challenge of Rating Structured Finance Assets

- These estimates of risk depend crucially on whether default correlations have been estimated correctly.
- One can explore the sensitivity of the original collateralized debt obligation and CDO-squared tranches to changes in default correlation for bonds within each collateralized debt obligation.
- The correlation in defaults for bonds belonging to different collateral pools remains fixed at zero.
- These illustrate the impact of either an error in the modeling assumptions or an unexpected realization of the default experience on the value of a 1 unit investment in each tranche.

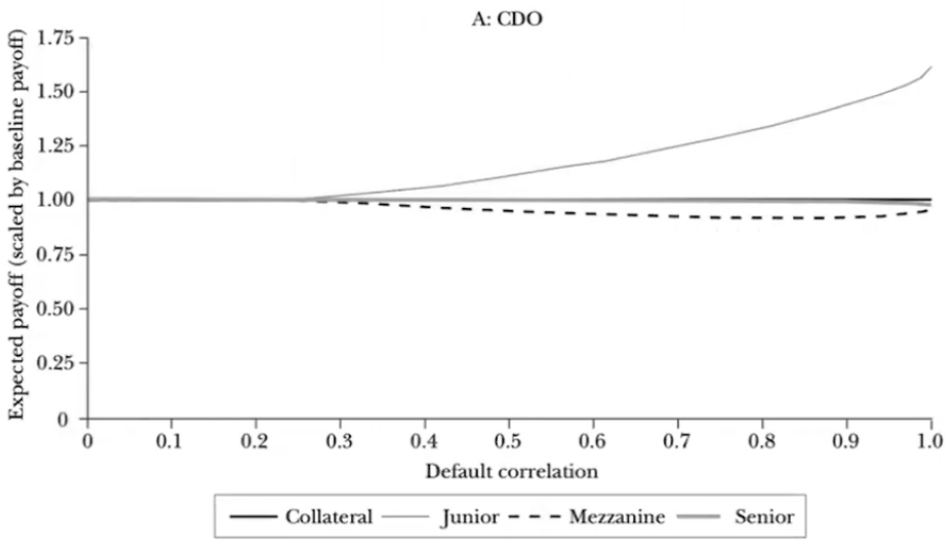


Table 3

Effect of Changes in Underlying Parameters on CDO and CDO² Tranche Ratings

		Final rating					
	Initial rating	*Default correlation (ρ)*			*Default probability (pD)*		
	(ρ = 20%, pD = 5%)	*40%*	*60%*	*80%*	*7.50%*	*10%*	*12.50%*
CDO							
Junior	NR	D	C	CC	NR	NR	NR
Mezzanine	BBB−	BB−	B+	B+	B+	CCC	CC
Senior	AAA	A+	BBB−	BB	AAA	A+	BBB−
CDO² ([6, 12])							
Junior	C	D	NR	NR	NR	NR	NR
Mezzanine	AAA	B+	C	CC	BBB−	NR	NR
Senior	AAA	AAA	AAA	AA+	AAA	AAA	B−

- A large fraction of collateralized debt obligations issued over the course of the last decade had subprime residential mortgage-backed securities as their underlying assets.
- Many of these residential mortgage-backed securities were themselves tranches from an original securitization of a large pool of mortgages, such that collateralized debt obligations of mortgage-backed securities are effectively CDO-squareds.

- As substantial lending to subprime borrowers was a recent phenomenon, historical data on defaults and delinquencies of this sector of the mortgage market was scarce.
- The possibility for errors in the assessment of the default correlations, the default probabilities, and the ensuing recovery rates for these securities was significant.
- Such errors, when magnified by the process of re-securitization, help explain the devastating losses some of these securities experienced.

GAs.

1) Suppose a one-year bond promises to pay 5,000 rupees a year from now. If price of the bond today is 4,000 rupees, then the interest rate on the bond is

- ○ 10%
- ○ 12.5%
- ○ 25%
- ○ 40%

$= \frac{5000 - 4000}{4000} = \frac{1}{4} = 25\%.$ $\left[\frac{\text{Amount} - P.}{P}\right]$

2) Consider the following order book at a time for Tata Motors.

Bid		Ask	
Price (Rs)	Quantity	Price (Rs)	Quantity
975.5	25	976	500
975	150	976.5	50
974.5	2000	977	100
974	80	977.5	75
973.5	500	978	5000

How many trades take place when a new buy order of 300 shares at Rs 976.3 is received?

Bid
300 shares at Rs 976.5
→ Then 300 shares will be given at the rate of 976

3) Consider the following order book at a time for Tata Motors.

Bid		Ask	
Price (Rs)	Quantity	Price (Rs)	Quantity
975.5	25	976	500
975	150	976.5	50
974.5	2000	977	100
974	80	977.5	75
973.5	500	978	5000

How many trades take place when a new sell order of 300 shares at Rs 975.3 is received?

Ask → 300 shares @ 975.3.
25 shares → At 975.3.

4) Efficient market hypothesis states that

- ○ A bubble is created when positive information is received about the market.
- ○ We can predict the timeline of a market crash.
- ✓ ○ Asset prices reflect all available information.
- ○ New information does not impact the asset prices.

Efficient Market Hypothesis says that market moves as per available info and "Bubble" cannot be predicted.

5) Compared to government bonds, stocks (shares) offer the holder

- ✓ ○ Higher risk and higher potential return
- ○ Higher risk and lower potential return
- ○ Lower risk and lower potential return
- ○ Lower risk and higher potential return

Bonds — lower risk, lower returns
Stocks — higher risk, higher return
higher the risk, higher the return.

6) Which of the following is/are the characteristics of a bond?

- [ ] If a bond has a high risk of default, investors will demand a lower interest rate ✗
- [ ] If a bond has a low risk of default, investors will demand a lower interest rate. (Yes)
- [ ] If a bond has a high risk of default, the bond price will be lower. Yes ✓
- [ ] If a bond has a low risk of default, the bond price will be lower. No.

7) Apple Inc. launched iPhone 16 series in the second week of September 2024. Which of the following statements are correct?

- [x] Investors can diversify the risk generated through this event.
- [ ] Investors cannot diversify the risk generated through this event.
- [x] This information is irrelevant for investor's portfolio valuation.
- [ ] This information impacts the valuation of investor's portfolio.

Risk → Idiosyncratic → can be diversified.
does not
or it impacts investor's portfolio.
Risk
Systematic → can't be diversified
2. impacts portfolio valued.
Idiosyncratic – opposite.

8) Which of the following are elements of a financial crisis?

- [x] A large decline in some asset prices
- [x] A decline in confidence in financial institutions
- [x] Insolvencies at financial institutions
- [ ] Stock market index reaching a new high level ✗

Reference for Q.9.

	Initial rating	Final rating: Default correlation (ρ)			Final rating: Default probability (pD)		
	(ρ = 20%, pD = 5%)	40%	60%	80%	7.50%	10%	12.50%
CDO							
Junior	NR	D	C	CC	NR	NR	NR
Mezzanine	BBB−	BB−	B+	B+	B+	CCC	CC
Senior	AAA	A+	BBB−	BB	AAA	A+	BBB−
CDO^2 ([6, 12])							
Junior	C	D	NR	NR	NR	NR	NR
Mezzanine	AAA	B+	C	CC	BBB−	NR	NR
Senior	AAA	AAA	AAA	AA+	AAA	AAA	B−

9) Consider the following statements regarding collateralized debt obligations discussed under simulation exercise. Select the correct statements

- [ ] As pairwise bond default correlation increases, the expected payoff of 'junior' tranches always decreases. → No. Yes. (normally) but not always
- [x] If default probability of bonds increases, the rating of 'senior' CDO may downgrade. Yes.
- [ ] As pairwise bond default correlation increases, the expected payoff of 'senior'tranches may decrease. No.
- [x] If default probability of bonds increases, the rating of 'junior' CDO may downgrade. yes.

10) Which of the following statements is correct? 6 p

- [x] The fundamental value of the shares in a firm is determined by expected future profits and systematic risk. Yes.
- [ ] If there is no new information regarding the future profitability or systematic risk of a firm, but its share price keeps rising, the fundamental value must be increasing. No, might be due to momentum based or speculative trading.
- [ ] Buying a share at a price above its fundamental value in the hope that someone else would buy it from you at an even higher price is guaranteed to lose money. Not True, may be momentum based. Risky but not guaranteed to lose.
- [ ] All investors always agree on the fundamental value of the shares in a firm. No.

11) Your portfolio consists of three blue bonds: each blue bond has a face value 100 and a probability of default p_b = 10%. The correlation between blue bonds is $\gamma = 0$. Through securitization, you are able to create three new bonds with face values: a green bond, a yellow bond and a red bond. When none of the bonds default all three new bonds are paid face value of 100. If any one of the blue bond defaults, the green bond and the yellow bond are paid the face value of 100 while the red bond suffers the loss and is paid 0. If two blue bonds default, only green bond is paid the face value, other two are paid 0. In case of all three blue bonds defaulting, none of the new bonds pays off. We are interested in calculating the fair value of the green bond, yellow bond and red bond. (Note: Write your answers up to three decimal points.)

What is the probability that all three blue bonds default? $= 0.1 \times 0.1 \times 0.1 = 0.001$

$= 0.1\%$.

What is the probability that none of the blue bonds default? $= 0.9 \times 0.9 \times 0.9 = 0.729$

$= 72.9\%$.

What is the probability that exactly one of blue bonds defaults?

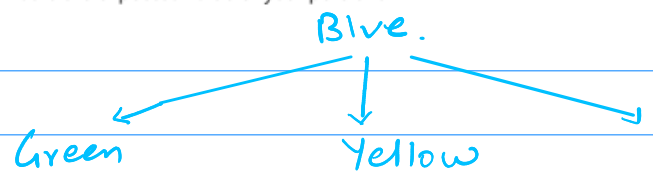
$= 3 \times 0.1 \times 0.9 \times 0.9 = 0.243 = 24.3\%$.

What is the probability that exactly two blue bonds default?

$= 3 \times 0.1 \times 0.1 \times 0.9 = 0.027 = 2.7\%$.

Check $0.1 + 72.9 + 24.3 + 2.7 = 73 + 27 = 100\%$ ∴ Correct.

15) Your portfolio has the following values: (i) 300 if none of the blue bonds default, (ii) 200 if exactly one blue bond defaults, (iii) 100 if exactly two blue bonds default, and (iv) 0 if all three blue bonds default. What is the expected value of your portfolio?



P (Green defaulting) = P (when all 3 default) = 0.1%.

Expected Payoff = 99.9% × 100 = 99.9.

P (Yellow defaut) = P(when all 3 default + when two default)

$= 0.1 + 2.7 = 2.8\%$. ∴ P(Not default) = 100 − 2.8

$= 97.2$

∴ E (payoff) = 97.2% 100 = 97.2.

P(Red defaults) = P (all 3 default + 2 defaul. + 1 default)

$= 0.1 + 2.7 + 24.3 = 27.1\%$.

P(Red not defaut) = 100 − 27.1 = 72.9%

E(Payoff) : 72.9.

Sum

∴ Expected Portfolio : 99.9 + 97.2 + 72.9.

= 270.0

16) What is the variance of your portfolio? $= E(X^2) - [E(X)]^2$.

$E(X^2) = P(\text{No one defaults}) \times 300^2 + P(\text{one defaults}) \times 200^2 + P(2 \text{ defaults}) \times 100^2$
$+ P(\text{all 3 defaults}) \times 0^2$.

$= 0.729 \times 90000 + 0.243 \times 40000 + 0.027 \times 10000$

$= 65610 + 9720 + 270 = 75600$

$\therefore$ Variance $= 75600 - (270)^2 = 75600 - 72900 = 2700$

What is the standard deviation of your portfolio?

$= \sqrt{2700} \approx 52 \quad (51.96)$

What is the expected payoff of a green bond across all states?

$= 99.9$

What is the expected payoff of a yellow bond across all states?

97.2

Red bond.

72.9

Key Concepts. 1. Calculate Probabilities of payoffs by.
— Green, Yellow & Red.
Green defaults only if all 3 fail.
Yellow defaults only if all 3/2 fail.
Red defaults if any 3/2/1 fail.